

**Intelligent Codebase and Domain Knowledge Management System: A
Generative AI Fine-Tuning Project**

Fine-Tuning GPT-2 on LangChain Docs with a Streamlit Web Interface

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Python

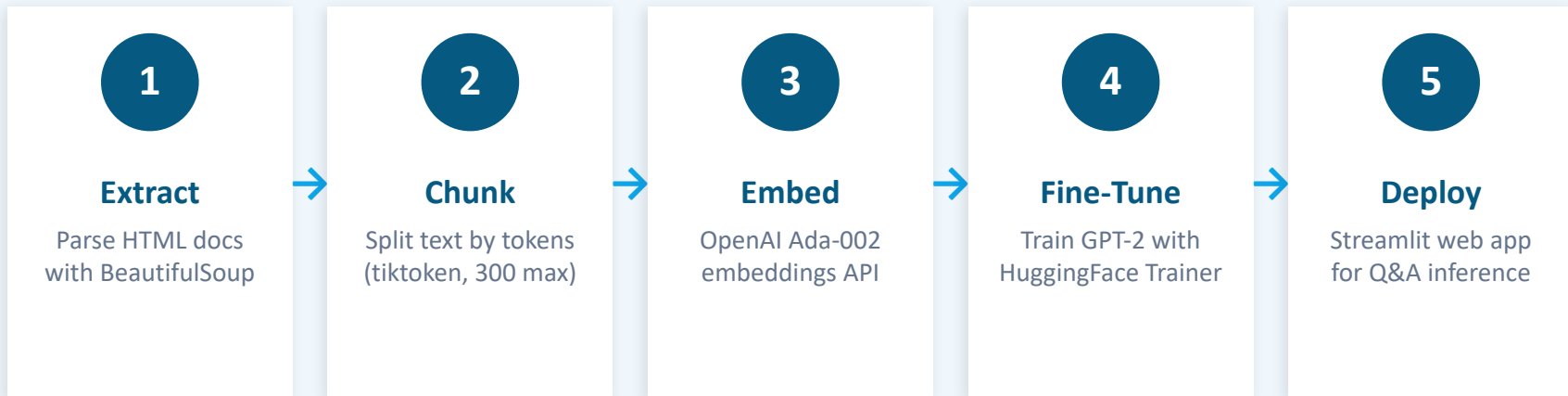
GPT-2

Streamlit

HuggingFace

Project Overview & Pipeline

This project builds a question-answering system for LangChain documentation. HTML docs are parsed, chunked, embedded via OpenAI, and used to fine-tune a GPT-2 model. The result is served through a Streamlit web app.



Key Libraries: BeautifulSoup | tiktoken | OpenAI API | HuggingFace Transformers | Streamlit

Fine-Tuning: Code Deep Dive

Data Preparation

```
extract_text_from_html()
```

Parses HTML docs via BeautifulSoup

```
split_text_by_tokens(max=300)
```

Uses tiktoken to chunk text for embeddings

```
calculate_embeddings_for_dict()
```

Calls OpenAI Ada-002 in batches of 1000

Training Configuration

Model: GPT-2 (124M params)

Epochs: 1 | Batch size: 8

Warmup: 500 steps

Weight decay: 0.01

Train/Val split: 80/20

Device: MPS (Apple Silicon)

Inference: generate_answer()

```
def generate_answer(question, model, tokenizer, max_length=150):  
    encoded = tokenizer(question, return_tensors='pt').to(device)  
    output = model.generate(input_ids, do_sample=True,  
                           top_k=50, top_p=0.95, temperature=0.7)  
    return tokenizer.decode(output[0], skip_special_tokens=True)
```

Streamlit App: Live Demo

App Features (app.py)

- Cached model loading

Uses `@st.cache_resource` for one-time load

- Auto device detection

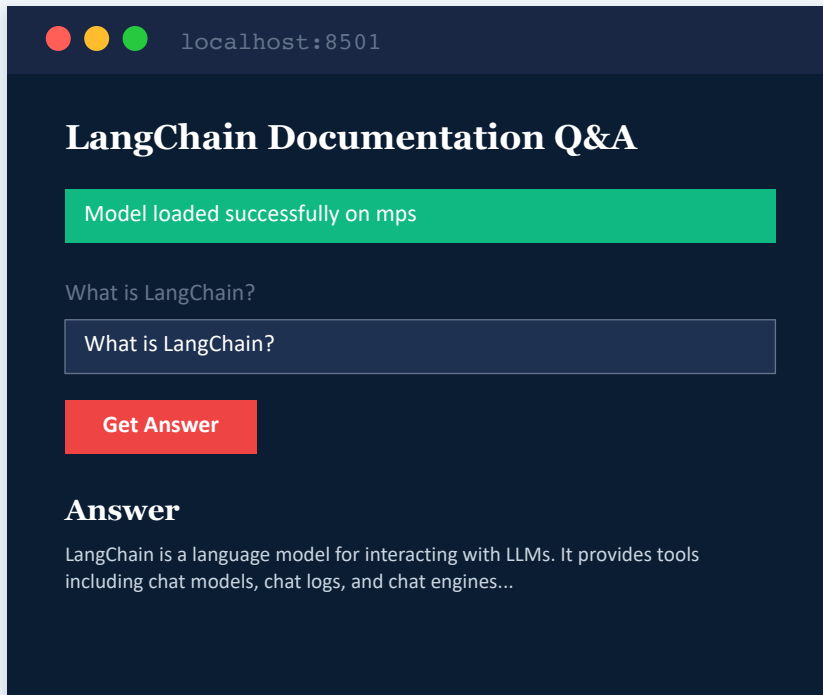
CUDA → MPS → CPU fallback chain

- Interactive sidebar controls

Max tokens slider (50-300)

Temperature slider (0.1-1.5)

- 5 sample questions pre-loaded
- Real-time text generation



Screenshot: Streamlit app running on localhost with MPS device

Results & Key Takeaways

Sample Q&A Results from Fine-Tuned Model

Question	Expected Answer
What is LangChain?	A framework for developing apps powered by LLMs
What is an LLMChain?	Simplest way to combine an LLM with a prompt template
How to install LangChain?	<code>pip install langchain</code>
What are Agents?	LLM deciding actions, observing, and repeating

Data Pipeline

HTML parsing + token-based chunking ensures consistent input for embeddings and fine-tuning.

Fine-Tuning

GPT-2 (124M params) trained in 1 epoch, showing even small models learn domain patterns.

Deployment

Streamlit provides an interactive UI with sidebar controls and cached model loading.